

Not All Channels Transfer: Coverage-Gated Fusion for Zero-Shot Cross-Country Detection of Online Information Operations

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Abstract—State-sponsored information operations (IO) are a growing threat to online platforms, and detectors must generalize to campaigns from countries never seen during training. The state-of-the-art graph foundation model IOHunter fuses text embeddings with a merged multi-relational similarity graph, but its zero-shot cross-country transfer degrades sharply on some targets. We trace this failure to coverage heterogeneity: the five behavioral sub-networks (co-retweet, co-URL, hashtag sequence, fast retweet, text similarity) cover wildly different fractions of accounts per country — for China, three of the five channels cover only 3–7% of accounts, so merging them floods the graph with isolated nodes that dilute the transferable coordination signal. We propose CS-DFA, a channel-specific architecture that runs one GNN per behavioral sub-network and fuses the resulting embeddings with coverage-gated attention, so each account aggregates only over channels it genuinely participates in. On the six-country IOHunter benchmark, CS-DFA raises average zero-shot test F1-macro from 0.775 (the strongest domain-alignment baseline) to 0.952 and improves on every country. Ablations isolate the mechanism: removing the coverage gate collapses performance to 0.686 — below the single-graph baseline — while removing channel-level CORAL alignment or the additive coverage prior changes results by at most 0.002. Splitting behavior into channels is thus not enough on its own, and explicit distribution alignment is unnecessary once low-coverage noise is gated out: what transfers across countries is the coordination structure itself.

Index Terms—information operations, social media mining, graph neural networks, zero-shot transfer, domain adaptation, coordinated behavior

I. Introduction

Coordinated information operations (IO) — often state-sponsored campaigns that deploy inauthentic accounts to manipulate public opinion — have been documented on every major social platform, targeting elections, geopolitical conflicts, and public-health discourse [1], [2], [3]. Policy bodies now treat foreign information manipulation and interference (FIMI) as a standing security threat [4], and platforms have released large archives of attributed IO accounts to support detection research [5].

A first family of detectors targets coordination: near-simultaneous retweeting, shared URL promotion, or synchronized hashtag use reveal accounts acting in concert [6], [7], [8], [9]. A second family frames the task as account classification with graph neural networks (GNNs) [10],

[11], [12], which have proven effective for inauthentic-account detection on social graphs [13]. IOHunter [14] combines both views into a graph foundation model [15]: it fuses Sentence-BERT text embeddings [16] with structural embeddings from a graph that merges five behavioral similarity networks (co-retweet, co-URL, hashtag sequence, fast retweet, tweet-text similarity), achieving state-of-the-art supervised performance on six country-level campaigns.

In deployment, however, the pressing regime is zero-shot: a new campaign emerges from a country for which no labels exist, and the detector must transfer from labeled campaigns of other countries. Under this protocol IOHunter’s performance drops steeply on several targets (e.g., F1-macro 0.58 on China), and standard domain-adaptation remedies — CORAL [17], adversarial alignment (DANN) [18], and graph-specific variants [19] — recover little (Section V).

Our diagnosis is that the bottleneck is not distributional misalignment but coverage heterogeneity. The five behavioral sub-networks cover very different fractions of accounts in each country (Table I): in the China campaign, the hashtag-sequence, fast-retweet, and text-similarity channels each cover only 3–7% of accounts, while co-retweet and co-URL cover 58% and 91%. An account absent from a channel is an isolated node in that channel’s graph; a GNN gives it no neighborhood aggregation, only a noisy pass-through of its own features. Merging all channels into one graph — as IOHunter and prior adaptation baselines do — lets this noise dilute the strong, highly transferable coordination signal carried by the high-coverage channels.

We therefore propose CS-DFA (Channel-Specific Decoupled Feature Alignment), which (i) keeps the five behavioral channels separate, running one GNN per sub-network, and (ii) fuses the per-channel embeddings with coverage-gated attention: a binary participation mask sets the attention score of every channel an account does not belong to to $-\infty$, so each account aggregates only over channels it genuinely participates in.

Our contributions are:

- A failure analysis of zero-shot cross-country IO detection showing that per-channel coverage sparsity, not language or feature shift, drives transfer failure on the hardest targets (Section III).
- CS-DFA, a channel-specific architecture with coverage-gated fusion (Section IV). On the six-country IOHunter benchmark it improves zero-shot F1-macro on every country, lifting the six-country average from 0.775 to 0.952 (Section V).
- A mechanism-isolating ablation with a negative result: disabling the gate collapses performance to 0.686 — below the single-graph baseline — while channel-level CORAL and an additive coverage prior are inert (≤ 0.002 change). Contrary to the premise of much domain-adaptation work, explicit distribution alignment is unnecessary here once low-coverage noise is gated out (Section V-C).

II. Related Work

A. Information Operations and Coordinated Behavior Detection

Early work characterized automated and state-sponsored accounts behaviorally [1], [20], [2]; content-based classifiers showed that IO posts are separable from organic activity within a campaign [3]. Platform data releases [5] enabled a line of work on coordination detection, which builds account-account similarity networks from co-actions — co-retweets, shared URLs, synchronized hashtags — and flags densely coordinated clusters [6], [7], [8]. Luceri et al. [9] showed that fusing several such similarity networks improves IO-driver classification. These similarity networks are exactly the sub-network channels our work treats as first-class citizens — but prior work merges them before learning, which we show is harmful under coverage sparsity.

B. GNNs and Graph Foundation Models for Account Classification

GNNs [10], [11], [12] are the dominant architecture for account classification on social graphs, e.g., relational GNNs for bot detection [13]. Graph foundation models [15] aim at transferable graph representations; IOHunter [14] instantiates this for IO detection by fusing SentenceBERT [16] text features with GNN structural features over the merged similarity network, and is the base model we extend. Our channel-specific view is related in spirit to multiplex/multi-relational GNNs, but our ablation shows the decisive ingredient is not the multi-channel split itself — which alone underperforms the merged graph — but gating fusion by coverage.

C. Domain Adaptation

Classic unsupervised domain adaptation aligns source and target feature distributions, via second-order statistics (CORAL) [17], [21], kernel two-sample discrepancies (MMD) [22], or adversarial objectives (DANN) [18];

UDAGCN adapts these ideas to graphs [19]. We evaluate CORAL and DANN under our protocol and find they barely move zero-shot performance, and that adding channel-level CORAL to CS-DFA is inert — evidence that for this task the transfer obstacle is structural noise, not distribution shift.

III. Problem and Motivating Analysis

A. Problem Formulation

Let $\mathcal{D} = \{D_1, \dots, D_6\}$ be country-level campaign datasets. Each D_k contains accounts V_k with binary labels (IO driver vs. organic), SBERT text embeddings X_k^{txt} , degree-based structural features X_k^{str} , and five behavioral sub-networks $\{G_k^c = (V_k, E_k^c)\}_{c \in \mathcal{C}}$ with channels $\mathcal{C} = \{\text{coRT}, \text{coURL}, \text{hashSeq}, \text{fastRT}, \text{tweetSim}\}$. In the zero-shot cross-country protocol, the model trains on the five source countries' labels only and is evaluated on the held-out target country's test split; target labels are never seen during training.

B. Coverage Heterogeneity

For channel c and country k , define coverage $\text{Cov}_k^c = |\{n \in V_k : \deg_{G_k^c}(n) > 0\}|/|V_k|$. Table I shows the six campaigns differ by up to two orders of magnitude in size and, critically, in per-channel coverage: co-retweet and co-URL consistently cover most accounts (24–93%), while hashtag sequence and fast retweet drop below 11% for half of the countries and text similarity swings from 3% to nearly 100% — for China all three cover only 3–7%. In a merged graph every account is implicitly present in all channels; the $\geq 93\%$ of Chinese accounts isolated in the sparse channels inject pure feature noise into message passing and dilute the coordination signal of coRT/coURL. This motivates keeping channels separate and masking each account's fusion to the channels it actually participates in.

IV. Method: CS-DFA

Figure 1 shows the architecture. CS-DFA has three stages: a decoupled multimodal projection inherited from our DFA baseline, five channel-specific GNNs, and coverage-gated attention fusion.

A. Decoupled Multimodal Projection (DFA backbone)

Following the decoupling philosophy of our strongest baseline DFA, text features X^{txt} (SBERT [16]) and structural features X^{str} (degree one-hot) are projected into a shared space by a cross-attention module in which each modality attends to the other; the two streams are then combined into multimodal node representations $H \in \mathbb{R}^{|V| \times d}$. This stage is identical across DFA and CS-DFA, so all gains reported later come from what follows.

TABLE I

The six IOHunter country campaigns [14]: graph statistics and per-channel coverage (% of accounts with ≥ 1 edge in that sub-network), computed from our processed graphs.

Country	#Accounts	#Edges	#IO drivers	IO %	Channel coverage (%)				
					coRT	coURL	hashSeq	fastRT	tweetSim
China	23,028	411,311	768	3.3	58.5	91.0	3.3	6.6	3.3
Iran	14,486	394,447	4,659	32.2	72.0	87.1	32.2	32.4	32.2
UAE	9,391	2,118,833	3,349	35.7	54.7	63.4	36.2	35.8	98.6
Cuba	20,247	4,737,799	461	2.3	23.9	49.0	4.4	2.5	100.0
Russia	716	10,431	275	38.4	63.6	93.4	38.4	38.4	38.4
Venezuela	5,021	56,741	533	10.6	69.8	91.9	10.6	10.9	10.6

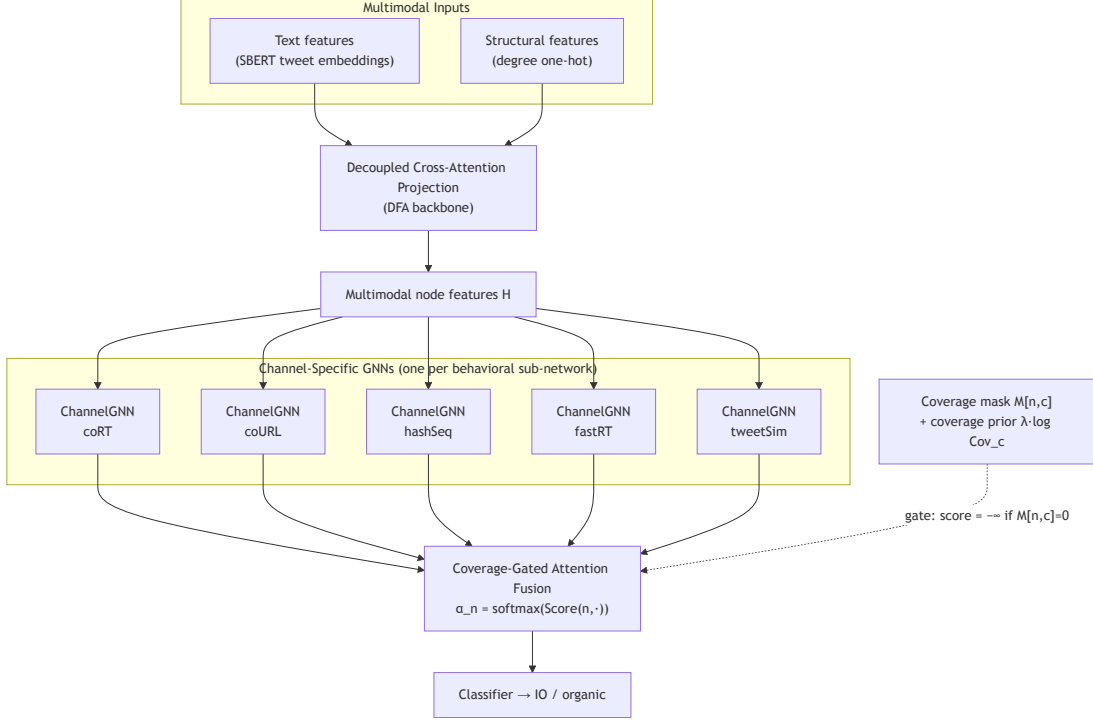


Fig. 1. CS-DFA architecture. Multimodal features (SBERT text + degree structure) are projected by a decoupled cross-attention module; one ChannelGNN runs per behavioral sub-network; coverage-gated attention fuses only the channels each account participates in.

B. Channel-Specific GNNs

Instead of a single GNN over a merged graph, CS-DFA runs one GNN per behavioral sub-network:

$$Z_c = \text{ChannelGNN}_c(H, E^c), \quad c \in \mathcal{C}, \quad (1)$$

yielding five channel embeddings $Z_c(n)$ per account n . Each channel’s message passing is thus confined to one behavioral semantics (e.g., pure co-retweet coordination), preventing cross-channel contamination.

C. Coverage-Gated Fusion

Let $M[n, c] \in \{0, 1\}$ indicate whether account n has at least one edge in channel c , and let Cov_c be the channel’s coverage in the account’s graph. A learned attention

function $\text{Att}(\cdot)$ scores each channel embedding, and the score is gated and biased:

$$\text{Score}(n, c) = \begin{cases} \text{Att}(Z_c(n)) + \lambda \log \text{Cov}_c, & M[n, c] = 1, \\ -\infty, & M[n, c] = 0, \end{cases} \quad (2)$$

$$\alpha_n = \text{softmax}(\text{Score}(n, \cdot)), \quad Z_{\text{fused}}(n) = \sum_{c \in \mathcal{C}} \alpha_{n,c} Z_c(n). \quad (3)$$

The hard gate ($-\infty$) guarantees an account fuses only channels it genuinely participates in: for China this removes the 3–7%-coverage noisy channels for $\sim 95\%$ of accounts, while for high-coverage countries it is nearly a no-op. The additive coverage prior $\lambda \log \text{Cov}_c$ gives structurally stable channels a head start ($\lambda=0$ recovers pure gating); our ablation shows it is in fact unnecessary.

TABLE II
Hyperparameters (defaults of the released runner).

Channel GNN	GraphSAGE, mean aggregation
Latent dimension d	128
Structural features	positional degree encoding
Dropout	0.2
Optimizer / learning rate	Adam / 10^{-2}
Max epochs / early-stop patience	1000 / 20 (target-val F1-macro)
Per-country training batch	128 accounts, class-stratified
Splits	5 random 60/20/20 splits
Loss	binary cross-entropy
Coverage prior λ	1.0
CORAL weight β (on coRT, coURL)	1.0
Tweet-similarity threshold	0.7
Random seed	12121995

A linear classifier on Z_{fused} predicts IO vs. organic.

D. Training Objective

We train with binary cross-entropy on source-country labels. (Focal loss [23] was tested and hurts extreme-imbalance targets such as Cuba by over-predicting the minority class, so we do not use it.) Optionally, a CORAL penalty [17] aligns second-order statistics of the stable channels’ embeddings (coRT, coURL) across source countries:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \beta \sum_{c \in \{\text{coRT}, \text{coURL}\}} \frac{1}{4d^2} \|C_{S_i}^c - C_{S_j}^c\|_F^2. \quad (4)$$

As Section V-C shows, this term can be dropped without measurable loss — we keep it in the full model only for completeness with the domain-adaptation literature.

V. Experiments

A. Setup

Data. The six country-level Twitter/X campaigns released with IOHunter [14] (Table I). Protocol. True zero-shot cross-country transfer: train on the five source countries, evaluate on the held-out target’s test split; repeat over five random splits and report mean $\pm$ std F1-macro. Model selection. Early stopping on the target validation split — shared by all compared methods; see Section V-F for why this inflates absolute (but not relative) numbers. Baselines. (i) IOHunter pre-trained zero-shot transfer, as reported in the original paper [14]; (ii) CORAL [17] and (iii) DANN [18] applied to the merged-graph model; (iv) DFA, our decoupled cross-attention alignment model and the strongest merged-graph baseline. All reproduced methods share the same features, splits, and selection protocol as CS-DFA. Implementation. PyTorch + PyTorch Geometric, trained on a single NVIDIA T4 (Colab); hyperparameters (the released runner’s defaults) are listed in Table II. Code: https://github.com/NelsonKCT/SMM_Final_Project.

B. Main Results

Table III reports zero-shot test F1-macro. CS-DFA improves on DFA for all six countries, lifting the average from 0.775 to 0.952 (+0.177). The largest gains are

precisely the low-coverage targets the analysis predicts: China +0.234 and Venezuela +0.230. Merged-graph domain adaptation (CORAL, DANN) performs at or below DFA, confirming that distribution alignment does not address the actual bottleneck. The only country where CS-DFA does not dominate every baseline is Cuba (0.888 vs. 0.899 for the paper-reported IOHunter number), the most extreme label-imbalance campaign (2.3% IO) — and paper-reported numbers were obtained under the original authors’ environment rather than our reproduced pipeline.

C. Ablation: the Gate Is the Contribution

Table IV disables each mechanism in turn (all variants share BCE loss and the training protocol). Removing channel-level CORAL changes the average by 0.002; removing the coverage prior ($\lambda=0$) changes it by 0.000. Removing the gate, however, collapses the average from 0.952 to 0.686 — a drop of at least 0.17 on every single country (Table V) — and lands below single-graph DFA (0.775). Two conclusions follow. First, splitting behavior into channels is not beneficial by itself; unmasked fusion lets isolated-channel noise pull every account’s representation around. Second, the entire improvement is attributable to coverage-gated fusion, a purely structural mechanism — explicit distribution alignment is unnecessary in this setting.

D. Mechanism Analysis

Attention distributions confirm the intended mechanism. With the gate off, per-account attention over the five channels is near-uniform on China ($\approx 0.21/0.25/0.16/0.18/0.19$), i.e., every account is pulled by its isolated-channel noise. With the gate on, attention concentrates on the channels the account truly belongs to (China: coRT 0.45, coURL 0.50; noisy channels $\approx 0.00x$), recovering the transferable coordination signal.

This pattern also argues against trivial label leakage as an explanation for the high scores. We audited the data path: target labels never enter training, structural features are pure node degree, and target splits are used only for evaluation and checkpoint selection. If the scores reflected a leak exploitable by any model, the ungated variant would also score high — it does not (0.686). The performance depends specifically on the gating mechanism.

E. Source-Data Efficiency

To measure how much labeled source data CS-DFA actually needs, we retrain with class-stratified subsamples of the source-country labels at 10% and 50% (drawn once per country with a fixed seed; target data and evaluation splits untouched), keeping the rest of the protocol fixed. Table VI shows that CS-DFA is strikingly data-efficient: with only 10% of the labeled source accounts it averages 0.958 F1-macro — statistically indistinguishable from its full-data 0.952, with per-country differences within one standard deviation — and its margin over DFA is stable

TABLE III

Zero-shot cross-country test F1-macro (mean $\pm$ std over five splits). [†]IOHunter numbers are as reported in the original paper [14]; all other rows are run under our unified pipeline. Best per column in bold.

Method	China	Iran	UAE	Cuba	Russia	Venezuela	Avg.
IOHunter (pre-trained) [†] [14]	0.581 $\pm$.059	0.728 $\pm$.014	0.839 $\pm$.059	0.899 $\pm$.054	0.798 $\pm$.019	0.910 $\pm$.011	0.793
CORAL [17]	0.585 $\pm$.031	0.705 $\pm$.016	0.878 $\pm$.016	0.848 $\pm$.079	0.823 $\pm$.030	0.768 $\pm$.031	0.768
DANN [18]	0.578 $\pm$.010	0.705 $\pm$.020	0.857 $\pm$.030	0.794 $\pm$.127	0.823 $\pm$.021	0.792 $\pm$.076	0.758
DFA (ours, merged graph)	0.603 $\pm$.030	0.706 $\pm$.009	0.880 $\pm$.018	0.856 $\pm$.091	0.836 $\pm$.024	0.768 $\pm$.031	0.775
CS-DFA (ours)	0.837 $\pm$.015	0.998 $\pm$.001	0.993 $\pm$.002	0.888 $\pm$.068	1.000 $\pm$.000	0.998 $\pm$.003	0.952

TABLE IV

Ablation on the fusion/alignment mechanisms (six-country average zero-shot test F1-macro).

Variant	Gating	Prior λ	CORAL	Avg. F1
CS-DFA (full)	on	1.0	channel	0.952
– CORAL	on	1.0	off	0.950
– prior	on	0.0	channel	0.952
– gating (bare backbone)	off	0.0	off	0.686
DFA (merged graph)	–	–	–	0.775

TABLE V

Per-country effect of removing the coverage gate (zero-shot test F1-macro).

	China	Iran	UAE	Cuba	Russia	Venez.	Avg.
CS-DFA (full)	0.837	0.998	0.993	0.888	1.000	0.998	0.952
w/o gate	0.510	0.671	0.817	0.619	0.767	0.733	0.686
Δ	−0.33	−0.33	−0.18	−0.27	−0.23	−0.27	−0.27

at every fraction (+0.200 at 10%, +0.186 at 50%, +0.177 at 100%). Because training draws a fixed 128-account stratified batch per country and epoch regardless of the pool size, the number of gradient updates is constant across fractions; what shrinks is the diversity of labeled accounts. Stability under a 10 $\times$ label reduction indicates that the gated coordination signal is learnable from a small set of labeled accounts: CS-DFA’s advantage does not depend on label volume.

F. Limitations and Threats to Validity

Target-validation model selection. Like the baselines we reproduce, early stopping selects the checkpoint maximizing target validation F1-macro. For several countries this validation score saturates quickly, so a higher-capacity model can be selected optimistically; absolute numbers (especially Russia/Iran/Venezuela $\approx$ 1.0) are likely inflated. All compared variants share the protocol, so relative conclusions — including every ablation finding — are unaffected. Developing a label-free model-selection protocol for the zero-shot setting (e.g., source-validation or last-epoch selection) is left as future work that would tighten the absolute claims. High-variance sparse slices. Per-channel test slices for the low-coverage channels carry $\pm$ 0.20–0.25 std; the headline average is carried by the high-coverage coRT/coURL slices. Benchmark scope. All six campaigns come from one platform and one release family; generalization to other platforms is untested.

TABLE VI

Zero-shot test F1-macro (six-country average) vs. fraction of labeled source-country data.

Method	10%	50%	100%
DFA	0.758	0.769	0.775
CS-DFA	0.958	0.955	0.952

VI. Conclusion

We showed that the key obstacle to zero-shot cross-country IO detection in multi-relational behavioral graphs is coverage heterogeneity, not distribution shift. CS-DFA keeps behavioral channels separate and fuses them with coverage-gated attention, improving zero-shot F1-macro on all six IOHunter campaign countries (average 0.775 $\rightarrow$ 0.952) and retaining its full-data performance with only 10% of the labeled source accounts. The ablation delivers a clean and somewhat contrarian message: channel splitting alone hurts, distribution alignment (CORAL) and coverage priors are inert, and the entire gain comes from letting each account fuse only the behavioral channels it genuinely participates in. What transfers across countries is coordination structure — provided the noise of low-coverage channels is gated out. Future work includes label-free model selection for the zero-shot setting, imbalance-aware loss routing for extreme-imbalance targets such as Cuba, and validation on non-Twitter platforms.

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